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Oefeningenreeks 5H nr. 1

$$\begin{cases} x(t) = t^2 \\ y(t) = 4t - t^3 \end{cases}$$

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$$V = \pi \int_0^2 (4t - t^3)^2 (t^2)' dt$$
$$= \pi \int_0^2 \left(16t^2 - 8t^4 + t^6\right) 2t dt$$
$$= \pi \int_0^2 \left(32t^3 - 16t^5 + 2t^7\right) dt$$
$$= \pi \left[8t^4 - \frac{8t^6}{3} + \frac{t^8}{4}\right]_0^2$$
$$= \pi \left(8 \cdot 2^4 - \frac{8 \cdot 2^6}{3} + \frac{2^8}{4}\right) - \pi \cdot 0$$
$$= \frac{64\pi}{3}$$

References